regulation 13.7.1.1 or certification pursuant to regulation 13.7.1.2 is not considered to be a substantial modification for the purpose of the application of regulation 13.2 of the annex. For recertification of

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regulation 13.7.1.1 or certification pursuant to regulation 13.7.1.2 is not considered to be a substantial modification for the purpose of the application of regulation 13.2 of the annex. For recertification of such an engine following a substantial modification, 2.5, 3.3 and, where that engine has multiple engine operating profiles, chapter 8 of this Code shall apply."
2 New paragraphs 1.3.21 to 1.3.37 are added as follows:
"1.3.21 Engine operational profile means a particular set of NOx influencing settings applied in the base emission control strategy which influences the NO x emission performance. Those settings may relate to, but are not limited to, fuel injection, inlet and exhaust valve operation, charge air management, exhaust bypass/wastegate or exhaust after-treatment controls and auxiliary control devices.

1.3.22 Multiple engine operational profiles means that more than one engine operational profile is available for selection on a marine diesel engine.

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OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 4
1.3.22 Multiple engine operational profiles means that more than one engine operational profile is available for selection on a marine diesel engine.
1.3.23 Auxiliary control device means a system, function or control strategy installed on a marine diesel engine that is used to protect the engine and/or its ancillary equipment against operating conditions that could result in damage or failure, or that is used to facilitate the starting of the engine. An auxiliary control device may also be a strategy or measure that has been satisfactorily demonstrated not to be a defeat device. An auxiliary control device includes any element of design that includes sensors, or other arrangements which, by an action of the control system, can activate, modulate, delay or deactivate the operation of any part of the base emission control system. Any device or strategy the activation of which causes a nonprogressive change in emissions is also an auxiliary control device. An auxiliary control device not declared at the time of the first certification of a marine diesel engine

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nonprogressive change in emissions is also an auxiliary control device. An auxiliary control device not declared at the time of the first certification of a marine diesel engine shall be considered a defeat device.
1.3.24 Defeat device means a device that measures, senses or responds to operating variables (e.g. engine speed, temperature, intake pressure or any other parameter) for the purpose of activating, modulating, delaying or deactivating the operation of any component or the function of the emission control system such that the effectiveness of the emission control system is reduced under conditions

encountered during normal operation, unless the use of such a device is substantially included in the applied emi ssion certification test procedures. An auxiliary control device accepted as part of the Administration's review of the NOx certification pack is not a defeat device.
1.3.25 Base emission control strategy means the emission control strategy active at any time an auxiliary control device is not active. It consists of any parameter, element of design, or operating control that is designed to modulate as a function of

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at any time an auxiliary control device is not active. It consists of any parameter, element of design, or operating control that is designed to modulate as a function of engine load and/or speed in a manner that affects the emission performance of the engine. The modulation of parameters is to be progressive and shall not result in a disproportionate change in emissions.
1.3.26 Rational emission control strategy means the base emission control strategy applied to a marine diesel engine which ensures that the emission values at the individual mode points as used to give the weighted specific emission value are representative of the emission values during normal operation of the engine.
1.3.27 Irrational emission control strategy means any strategy or measure that, when a marine diesel engine is operated under normal conditions of use, reduces the effectiveness of an emission control system to a level below that expected from the applicable emission test procedures.
1.3.28 Not-to-exceed emission limit value means the maximum permitted NO x emission value at a given operating condition as determined in accordance with 3.3

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applicable emission test procedures.
1.3.28 Not-to-exceed emission limit value means the maximum permitted NO x emission value at a given operating condition as determined in accordance with 3.3 of this Code within the not-to-exceed zone of the engine.
1.3.29 Not-to-exceed zone means the power or torque and speed area of a marine diesel engine within the limit area of the not-to-exceed zone as declared by the applicant that the engine is certified to operate within under steady -state conditions. In the case of the C1 cycle, as given by 3.2 of this Code, the not-to-exceed zone corresponds to the whole of the limit area of the not-to-exceed zone.
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 5 -
1.3.30 Limit area of the not-to-exceed zone means the power or torque and speed boundaries of the not-to-exceed zone at and above 25% rated power for all test cycles as given by 3.2 of this Code except for the C1 cycle where it is at and above 50% engine load.
1.3.31 Point emission value means the NO x emission value expressed in terms

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as given by 3.2 of this Code except for the C1 cycle where it is at and above 50% engine load.
1.3.31 Point emission value means the NO x emission value expressed in terms

of g/kWh at the reference conditions of humidity and temperature given by this Code at a particular power or load and speed point.
1.3.32 NOx certification pack means the package of information supplied by the applicant to the Administration as required to be submitted by 2.5 and 3.3 of this Code.
1.3.33 Propulsion engine means a marine diesel engine that is used for direct or indirect propulsion. A propulsion engine may additionally perform non -propulsion duties during or separately to propulsion duties.
1.3.34 Non-propulsion engine means a marine diesel engine that is not a propulsion engine. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.
1.3.35 Constant-speed engine means a marine diesel engine that is limited to constant-speed operation.
1.3.36 Constant-speed engine operation means a marine diesel engine regulated by a speed control device that automatically controls the operator demand to maintain

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constant-speed operation.
1.3.36 Constant-speed engine operation means a marine diesel engine regulated by a speed control device that automatically controls the operator demand to maintain engine's nominal speed across the load range.*
Additionally, an idle speed setting may be provided that can be used during start -up or shutdown.

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* In service, such a speed control device may either maintain a fixed speed or a load dependent speed such that at maximum load the speed could be up to around 10% lower than at zero load.
1.3.37 Variable-speed engine means an engine that is not a constant -speed engine."
Chapter 2 Surveys and certification
3 A new section 2.5 is added as follows:
"2.5 Rational emission control strategy
2.5.1 In addition to 2.2, the requirements of this section shall apply.
2.5.2 A rational emission control strategy shall be applied to each marine diesel engine across the whole of its operating load and speed range. The means by which that is achieved shall be documented by the applicant to the Administration within a NOx certification pack. The information included in that pack shall be such as to

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that is achieved shall be documented by the applicant to the Administration within a NOx certification pack. The information included in that pack shall be such as to demonstrate to the satisfaction of the Administration that a rational emission control strategy is applied during normal operation of the engine.
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 6
2.5.3 For an engine where one or more auxiliary control devices are applied, each of those shall be declared to the Administration within the NO x certification pack irrespective of whether those operate under steady -state or transient conditions. An auxiliary control device which is not so declared shall be considered a defeat device and hence invalidate the NO x certification of an engine to which such an undeclared device is applied.

2.5.4 For screening of the base emission control strategy, the NO x certification pack shall include:
.1 a list of all NO x emission influencing setting and operating values controlled by an engine's base emission control strategy, for

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2.5.4 For screening of the base emission control strategy, the NO x certification pack shall include:
.1 a list of all NO x emission influencing setting and operating values controlled by an engine's base emission control strategy, for
example fuel injection, inlet and exhaust valve operation, charge air management, exhaust bypass/wastegate or exhaust after-treatment controls;
.2 a record of the reference values for the settings and operating values identified in 2.5.4.1 at each of the mode points of the applicable test cycle;
.3 documentation that whenever the engine is operating between two mode points as identified in 2.5.4.2, the emission control strategy interpolates progressively between the mode points;
.4 documentation to show that, along lines of constant power and varying speed from the line between the mode points to the limit